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INTERIM FEDERAL SPECIFICATION

MEATLOAF WITH MUSHROOM GRAVY, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers meatloaf with mushroom gravy thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495	- Coatings, Exterior, for Tinned Food Cans
PPP-B-636	- Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specification for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars used in the Preparation of Thermostabilized Foods for the Armed Forces.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture (USDA)

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United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Grades of Nonfat Dry Milk

(Application for copies should be addressed to the Director, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Grades of Tomato Paste
U.S. Standards for Grades of Canned Mushrooms

(Application for copies should be addressed to the Director, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Regulations Governing the Inspection of Eggs and Egg Products

(Application for copies should be addressed to the Director, Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S Government Printing Office, Washington, DC 20402.)

Cereal Panel Odor and Flavor Evaluation System

(Application for copies should be addressed to the Northern Regional Research Center (ARS-USDA), 1815 North University, Peoria, IL 61604.)

Environmental Protection Agency (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Dehydrated Onion and Garlic Association (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association (ADOGA), 251 Kearny Street, #410, San Francisco, CA 94108.)

National Academy of Sciences

Food Chemicals Codex

(Application for copies should be addressed tot he National Academy Press, 2101 Constitution Avenue, NW, Washington, DC 20418.)

Association of Official Analytical Chemists (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1 and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of insect or rodent infestation, extraneous material, off-odors, off-flavors, and off-colors.

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3.2.1 Beef. Beef shall be the only meat ingredient used and shall be from steers, heifers, or cows and shall be derived from any combination of the following recognizable primal and sub-primal cuts: square-cut chucks, ribs, trimmed full loins, trimmed short loins, strip loins, sirloins, top sirloin butts, rounds, shoulder clods, chuck rolls, ribeyes, top rounds, knuckles, and bottom rounds. Recognizable cuts are those which, when compared to IMPS cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. All suitable lean meat shall be used except tenderloins may be excluded at the option of the contractor. The beef shall be in the fresh chilled state and shall be in excellent condition, i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class and cut of meat, and typical of meat which has been properly stored and handled; cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to dehydration, aging, or microbial activity. The fat shall show not more than a very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable. Also, meat shall show no evidence of freezing, defrosting or mishandling.

3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, heavy connective tissue, etc. The boneless, trimmed beef shall meet the limitations specified in tables I and II.

3.2.1.2 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the same day as boning and trimming shall be maintained in the temperature range of 28° to 50°F (inclusive).

b. Holding in the fresh chilled state not more than 4 days is permitted provided that the beef is maintained in the temperature range of 28° to 40°F (inclusive).

c. Holding raw beef in the frozen state is permitted provided that the beef is:

-placed in freezer within 4 hours after certification and frozen to 0°F within 72 hours

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-stored at 0°F or lower
-held for not more than 60 days
-protected from freezer deterioration and damage
-stored in containers that are adequate to maintain product excellence
-maintained without defrosting until time of processing into the finished product

3.2.2 Eggs, whole. The whole eggs may be liquid or frozen and shall have been processed in accordance with 7 CFR part 59, Regulations Governing the Inspection of Eggs and Egg Products. The whole eggs shall have been prepared from egg yolks and egg whites in their natural proportions as broken directly from the shell eggs as evidenced by a USDA Egg Products Inspection Certificate. The whole eggs shall contain no additives and shall be in satisfactory condition, free of objectional odors, such as musty, sour, or fruity, and free from foreign material. Liquid egg products shall be held at a temperature of 40°F or lower and shall not be held for more than 72 hours from the time of pasteurization until the start of formulation of the eggs into product. Frozen whole eggs shall be held at 10°F or lower and used within 120 days. Thawed previously frozen egg products shall be held at a temperature of 40°F or lower and shall not be held for more than 24 hours prior to use in product.

3.2.3 Tomato paste. Tomato paste shall be U.S. Grade A or better of the U.S. Standards for Grades of Tomato Paste. See formulation (3.3) for required soluble solids content.

3.2.4 Bread crumbs. Bread crumbs shall be made from fresh white bread only. The bread crumbs shall be coarse ground, free-flowing and free from burnt, rancid, moldy or musty odors or flavors.

3.2.5 Protein, soy, powdered. The powdered soy protein shall be derived from either soy protein concentrate or soy protein isolate and shall have the following characteristics:

Chemical characteristics

a. Protein (N x 6.25) m.f.b.	69.0% (min)
b. Moisture	7.0% (max)
c. Fat (AOCS Method EE3-49)	2.0% (max)
d. Ash	7.0% (max)
e. pH	6.8 to 7.1
f. Nitrogen solubility index	50.0 (min)

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Physical requirements

Particle size. At least 99.0 percent shall pass through a U.S. Standard No. 100 sieve.

Sensory characteristics

a. Shall receive a flavor score of 6.0 or higher when tested in accordance with the provisions of the U.S.D.A. Northern Regional Research Center Odor and Flavor Intensity Scoring Method of the Cereal Panel Odor and Flavor Evaluation System.

b. Shall have no added flavor.

3.2.6 Onion, dehydrated, chopped. The dehydrated, chopped onion shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.7 Celery, sliced, freeze-dried. The freeze-dried celery shall be obtained from pascal type celery. The celery shall be cut transversely into 1/8 to 3/8 inch slices (with no leaf cuts), sulfited, freeze-dried, or partially air-dried and freeze-dried. The celery shall be free flowing and have an odor and flavor typical of fresh celery. The product shall reconstitute to a pale to dark green color. The moisture content shall be not greater than 4 percent and the sulfite level shall be in the range of 500 to 1500 ppm as SO_2.

3.2.8 Onion powder. Onion powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.9 Salt. Salt shall be noniodized, white, refined, sodium chloride with or without anitcaking agents.

3.2.10 Peppers, green, sweet, freeze-dried. The peppers shall be 1/4 inch dices of sweet bell peppers which have been sulfited and dehydrated by freeze-drying or partially air-dried followed by freeze-drying. The dices shall be free of seeds, seed cells and diseased or rotted areas. The pepper dices shall have a fresh, clear typical sweet green pepper flavor and odor and green with minimal brownish color. The moisture content shall be not greater than 4 percent and the sulfite level shall be in the range of 500 to 1500 ppm as SO_2.

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3.2.11 Sugar, granulated. Sugar shall be refined, sucrose, granulated cane or beet, or a combination of both and shall comply with MIL-STD-900.

3.2.12 Monosodium glutamate (MSG). Monosodium glutamate (MSG) shall comply with the Food Chemicals Codex.

3.2.13 Garlic powder. Garlic powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.14 Pepper, black, ground. The ground black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic penetrating odor, a hot biting pungent flavor and a light gray to speckled black-gray color. The black pepper shall contain not less than 2.0 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.15 Celery seed, ground. Ground celery seed shall be prepared from the seed of Apium graveolens L. The ground celery seed shall be light to rich brown in color and possess a characteristic celery odor and flavor with a warm, slightly bitter aftertaste. Celery seed shall contain not less than 2.0 mL of volatile oil per 100 grams of ground celery seed. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 55 sieve.

3.2.16 Allspice, ground. Ground allspice shall be derived from dried, nearly ripe fruit of Pimenta officinalis Lindl and shall have a fragrant, clove-like aroma, strongly aromatic, pungent, clove-like flavor and a dark reddish-brown color. Allspice shall contain not less than 3.0 mL of volatile oil per 100 grams of ground allspice with 80.0 percent eugenol as a principal constituent. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 25 sieve.

3.2.17 Bay leaves, ground. Ground bay leaves shall be derived from dried leaves of Laurus nobilis L. The bay leaves shall possess a pleasant aromatic odor and pungent, mildly bitter flavor with a pale green to yellow green color. A minimum of 95 percent shall pass through a U.S. Standard No. 30 sieve. Volatile oil content shall be not less than 1.0 mL of volatile oil per 100 grams of ground bay leaves.

3.2.18 Mushrooms, canned, stems and pieces. Canned mushroom stems and pieces shall be U.S. Grade B or better in accordance with the U.S. Standards for grades of Canned Mushrooms.

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3.2.19 Starch, modified, high opacity. The high opacity starch shall be white, odorless, finely pulverized modified maize food starch for use in canned foods. The modified starch shall demonstrate initial viscosity development in the temperature range 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperature. The starch shall resist breakdown at low pH, under shear stress and conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.20 Soup stock, dehydrated. The soup stock shall be a light brown to tan colored, free flowing powder derived from soup stock. The soup stock shall be made from USDA inspected beef or pork bones with adhering meat or a combination of the two and shall have a minimum of 93 percent total solids and a maximum of 1 percent fat.

3.2.21 Oil, vegetable. The vegetable oil shall have a stability of 15 hours AOM (minimum), a free fatty acid content not to exceed 0.05 percent by weight (as oleic acid), a peroxide value not to exceed 1.0 mg of peroxide per kilogram of fat, a moisture/volatile matter not to exceed 0.06 percent by weight, and shall pass the cold test of 5.5 hours (minimum).

3.2.22 Color, caramel. Caramel color shall meet the requirements of the Food Chemicals Codex.

3.2.23 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.24 Water. Water used for formulation, ice making, and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.2.25 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Preparation of the meatloaf mixture. The meatloaf shall be prepared from the following ingredients:

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Ingredient	Percent by weight
Beef, ground	70.94
Water	11.45
Eggs, whole	4.00
Tomato paste (24% natural tomato soluble solids) 1/	3.00
Bread crumbs	2.00
Protein, soy, powdered	2.00
Onion, dehydrated, chopped	2.00
Celery, sliced, freeze-dried	1.00
Onion powder	1.00
Salt	0.85
Peppers, green, sweet, freeze-dried	0.50
Sugar, granulated	0.50
MSG	0.50
Garlic powder	0.08
Pepper, black, ground	0.07
Celery seed, ground	0.05
Allspice, ground	0.03
Bay leaves, ground	0.03

1/ If the specified tomato solids are not available, the tomato solids that are used must be calculated to bring the solid level to that specified and water adjusted accordingly.

The meatloaf mixture shall be prepared as follows:

a. The chilled boneless beef shall be ground once through a 1-inch plate. Time from grinding to mixing shall not exceed 4 hours. The beef shall be maintained in the temperature range of 28° to 40°F during the time period between grinding and mixing.

b. Bread crumbs, salt, black pepper and garlic powder shall be added to the coarse ground beef, thoroughly mixed, and then the mixture ground through a 1/4-inch plate.

c. The remaining ingredients shall be mixed together to form a slurry.

d. The ground beef mixture and the slurry from steps b and c shall be thoroughly mixed together in a mechanical mixer.

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3.3.2 Stuffing and forming. The meatloaf mixture shall be immediately mechanically stuffed into number 6-1/2 prestuck E-Z-peel casings. The stuffed meatloaf logs shall be formed by pressing between spring loaded screens separated by 1-7/8 inch spacers, such that rectangularly shaped loaves are formed that are 1-5/8 to 1-7/8 inches in height.

3.3.3 Cooking and cooling. The stuffed meatloaf loaves shall not exceed 40°F between stuffing and cooking and shall be in the cooking process within 4 hours after being stuffed. The loaves shall be dry heat cooked at 190° to 200°F until heated to an internal temperature of 160° to 170°F. The cooked loaves shall be cooled immediately to not more than 40°F without freezing. The loaves shall not be held longer than 48 hours from time of removal from the cooking process until filling into the tray cans.

3.3.4 Slicing. The cooked cooled meatloaves shall be uniformly sliced into slices, 40 of which fit snugly in a tray pack can.

3.3.5 Gravy preparation. The gravy shall be prepared from the following ingredients:

Ingredient	Percent by weight
Water	72.83
Tomato paste (24% soluble solids)	7.50
Mushrooms, canned	6.00
Starch, modified, high opacity	5.50
Soup stock, dehydrated	3.50
Oil, vegetable	1.00
Onion powder	1.00
Salt	0.85
Sugar, granulated	0.50
MSG	0.50
Color, caramel	0.35
Lecithin	0.25
Pepper, black, ground	0.07
Garlic powder	0.06
Allspice, ground	0.03
Bay leaves, ground	0.03
Celery seed, ground	0.03

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The gravy shall be prepared as follows:

a. The mushrooms and brine from the cans shall be blended together to produce a smooth, homogenized mixture.

b. A slurry shall be made with the starch and part of the water.

c. The mushroom mixture and the remaining ingredients shall be mixed together and heated to approximately 165° to 170°F. The slurry shall be added and the final mixture heated to 165° to 170°F and held for approximately 5 minutes at this temperature.

d. The volume of the final mixture shall be adjusted to compensate for evaporation loss during heating.

e. The gravy shall be filled within 4 hours after cooking, and held at a temperature of not less than 150°F during this interval.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. Not less than 40 uniform slices of meatloaf shall be placed on edge in two snug rows into the tray pack can and the can (see 5.1.1) immediately filled with gravy to not less than 102 ounces of product.

b. The temperature of the meatloaf at time of filling shall be 28° to 40°F.

c. The temperature of the gravy at time of filling shall be not less than 150°F.

d. Each can shall be hermetically sealed with not less than 10 inches (Hg) vacuum immediately after filling.

e. The filled and sealed tray pack cans shall be in the retort process within one hour after filling.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at 240°F or below until a sterilization value (F_o) of not less than 6 has been achieved.

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3.6 Finished product requirements. The finished product shall comply with the following requirements.

a. There shall be no foreign material, such as, but not limited to dirt, insect parts, hair, wood, paper, paint, glass or metal.

b. There shall be no foreign odor or flavor, such as, but not limited to burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. There shall be not less than 40 uniform intact slices per tray pack can.

e. The drained weight of 40 slices in an individual can shall be not less than 66 ounces.

f. The average net weight shall be 102 ounces minimum.

g. No individual can shall contain less than 100 ounces of product.

h. Texture of the meat loaf shall not be dry, rubbery, and/or mushy.

i. Gravy shall be light brown to dark brown in color.

j. The gravy shall have a smooth (not lumpy) consistency without being excessively thick or thin.

k. The average fat content of the finished product shall be not greater than 8.0 percent and no individual tray pack shall have a fat content which is greater than 10.0 percent.

l. The salt content of the finished product shall be not less than 0.5 percent nor greater than 1.25 percent in any individual can.

m. Slices shall be uniform in shape and size.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

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3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this specification.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or other applicable purchase document.

4.4.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts. Any nonconforming beef shall be rejected.

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4.4.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size expressed in terms of pounds shall be declared to the Agricultural Marketing Service (AMS) agent by the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirements. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or presence of any off-condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

TABLE I. Sampling plan for boned and trimmed beef

Lot size (pounds)	Sample size (No. of sample units)	Defect categories			
		Major		Minor	
		AC	RE	AC	RE
500 or less	13	1	2	3	4
501 to 1200	20	2	3	5	6
1201 to 3200	32	3	4	7	8
3201 to 10000	50	5	6	10	11
10001 to 35000	80	7	8	14	15
Reinspection lots of 10001 to 35000	125	10	11	21	22

TABLE II. Boned and trimmed beef defects

Category		Defect 1/ 2/
Major	Minor	
151		Presence of lymph gland measuring 1.0 inch or more in any dimension.

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TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
	201	Presence of lymph gland measuring 0.5 inch or more but less than 1.0 inch in any dimension.
152		Presence of blood clot measuring 1.0 inch or more in any dimension.
	202	Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension.
153		Presence of bruise measuring 1.0 inch or more in any dimension.
	203	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension.
154		Presence of bone measuring 0.3 inch or more in any dimension.
155		Presence of cartilage measuring 0.5 inch or more in any dimension.
156		Presence of backstrap or other ligament measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	204	Presence of backstrap or other ligament measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
157		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 2.0 square inches or more.

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TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
	205	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 0.5 square inches or more but less than 2.0 square inches.
	206	Presence of heavy connective tissue on lower edge of short plate or flank.
	207	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension.
	208	Presence of membranous portion of diaphragm or membranous covering from skirt, flank or abdominal section of short plate measuring 3 square inches or more.
158		Presence of knuckle cover.
159		Presence of kidney, pizzle eye, pre-pubic tendon, thymus gland or hanging tender measuring 1.0 inch more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	209	Presence of kidney, pizzle eye, pre-pubic tendon, hanging tender or thymus gland measuring less than 1.0 inch in longest dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	210	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more.

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TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
	211	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface.
	212	Presence of dehydrated surface measuring 1.0 square inch or more.
	213	Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more.
	214	Presence of exposed major blood vessel measuring 1.0 inch or more in any dimension.
	215	Presence of cod, udder, kidney, or pelvic fat.

1/ Acceptability of product with respect to the presence of foreign material (for example: glass, dirt, insect parts, hair, wood, metal) shall be made by a Meat and Poultry Inspection Operations employee. The presence of any product not in excellent condition shall cause rejection of the involved lot.

2/ Evidence of freezing or defrosting or presence of any off-condition (including but not restricted to being off-odor, slightly sticky, gassy, rancid, sour, or showing discoloration) shall cause rejection of the involved lot.

4.4.3 Ingredient and component examination. Conformance of ingredients and components to identity, condition and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be evident by examination of pertinent labels, markings, U,S. Grade Certificates, certificates of analyses or other such supplier-furnished valid documents acceptable to the inspection agency. In addition, prior to use, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

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4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 102 ounces or if the net weight of any individual can is less than 100 ounces, the lot shall be rejected.

4.6.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table III. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated at least 30 minutes in 160° to 180°F water, opened, and inspected for the defects listed in table IV.

TABLE III. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (No. of cans)	Cumulative sample	Acceptance number	Rejection number
0-3200	8		0	2
	8	16	1	2
3201-35000	13		0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

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b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but are less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE IV. Product defects 1/ 2/

Category	Defect
Major	
151	Less than 40 intact slices.
152	Drained weight of 40 meatloaf slices in an individual can is less than 66 ounces. 3/
153	Gravy not light brown to dark brown in color.
154	Gravy consistency not smooth (lumpy) or is excessively thick or thin.
155	Texture of meat loaf slices dry, rubbery, or mushy.
156	Slices not uniform in shape or size.

1/ The presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale, etc.) or foreign color shall be cause for rejection of the lot.

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2/ Product not equal to or better than the approved preproduction sample in palatability or overall appearance shall be cause for rejection of the lot.(This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ To determine drained weight, the free liquid shall be poured off and the remaining contents shall be poured into a flat bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated such as to liquify rendered fat without breaking the meatloaf slices. The contents shall then be poured into a U.S. Standard 1/4-inch sieve in a manner that will distribute the product over the sieve without breaking the meatloaf slices. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at an approximate 45 degree angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weights shall be reported to the nearest 0.1 ounce.

4.6.3 Fat and salt content testing. Nine tray pack cans shall be selected from the lot and distributed as follows:

-Three for laboratory analysis.
-Three for submission to the contractor.
-Three for retention by an AMS agent as a reserve sample.

The three tray packs for laboratory analysis shall individually be tested for fat and salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists: Chapter: Meat and Meat Products; Section, Meat, except that preparation of the sample shall be as follows: The three unopened tray packs shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The lot shall be rejected if:

a. The average fat content of the three cans is greater than 8.0 percent.

b. The fat content in an individual can is greater than 10.0 percent.

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c. The salt content in any individual can is less than 0.5 or greater than 1.25 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the fat or salt content requirement. When the reserve sample is analyzed, the analyses for both fat and salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lot.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that the inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack marking missing, incorrect, or illegible; Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Reinforcement with nonmetallic strapping or tape is not used; (4) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above top edge of box; (5) Length or width of pads less than box dimensions by more than 1/8 inch; (6) Arrangement or number of cans not as specified.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 102 ounces of the food product, sealed, and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-plated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated.

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Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of grade V3c fiberboard. The pads shall be placed between the cans and on the top and bottom of the stacked cans. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). Strapping shall be limited to non-metallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimension of the box and shall be fabricated of the same material as the box.

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5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.7 after submerging the can in boiling water for 30 minutes. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 35 minutes. Avoid overheating to the point where internal pressure is evident (can shows evidence of bulging). Remove from water and open lid for serving. Caution: Use care when opening as pressure may have been generated with the can.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F. WARNING: Do not place closed can in oven as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 20 servings of 2 slices each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary markings in capital letters larger than other markings shall be included:

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CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.3 and 6.3).
c. Provisions for approved preproduction sample (see 3.7 and 6.3).
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Type and class of unit load when unit loading is required (see 5.3).

6.2 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article shall be a preproduction sample. The contracting officer should include specific instructions in all procurement documents regarding arrangements for selection, inspection, and approval of the first article.

Preparing activity:

Army - GL

Project No. 8940-A470

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
EE-M-002161A(Army-GL)	Meatloaf with Mushroom Gravy, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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